FORDHAM HINF 6117 • SPRING 2026

# ChatEHR Main Project Guide

Teams of 3-4 will use a realistic synthetic EHR to compare AI models and settings, document evidence in the Team Notebook, and defend one recommended configuration for clinical use.

EXPECTED EFFORT

**4-8 hours per person**

Spread across configuration experiments, benchmarking, observability review, and final presentation prep.

## What You Must Learn to Explain

- Why different models produce different clinical answers.
- How temperature, context size, and response style affect reliability.
- When RAG helps and when it adds noise.
- How input tokens, output tokens, latency, and cost change by setup.
- How to assess safety, bias, health equity, and deployment risk.

## Core Team Roles

- **Clinical lead:** checks chart evidence and patient safety.
- **Experiment lead:** changes one variable at a time.
- **Observability lead:** tracks tokens, cost, latency, and retrieved context.
- **Equity and governance lead:** documents bias, language, access, and fairness concerns.

## Required Workflow

1. Create or join your team and save at least 3 configurations.
2. Run guided comparisons on the same patient and prompt while changing one setting at a time.
3. Record hypotheses and results in the Team Notebook after each experiment.
4. Complete the practice benchmark, then one checkpoint benchmark by March 24, 2026.
5. Review observability data: tokens, cost, latency, and RAG evidence.
6. Complete the final benchmark and submit the in-app reflection before the presentation.

## What to Compare

- Small vs larger models on note synthesis and medication reconciliation.
- Low vs high temperature on consistency and hallucination risk.
- Minimal vs broad context on accuracy, privacy, and token cost.
- RAG off vs on for guideline-grounded prompts.
- Different response styles for clinician-facing vs patient-facing output.

## Project Milestones

| | |
|---|---|
| **WEEK 1** | Onboard, assign roles, save baseline configs, and complete 2 guided experiments. |
| **WEEK 2** | Run practice benchmark, review observability, and capture at least 4 notebook entries with evidence. |
| **MAR 24** | Checkpoint benchmark due. Be ready to show an early recommendation and explain tradeoffs. |
| **APR 7** | Final benchmark, final reflection, and team presentation due. |

## Minimum Deliverables

3 saved configurations, 4 notebook entries, 1 practice benchmark, 1 checkpoint benchmark, 1 final benchmark, 1 final reflection, and a short team presentation defending your recommended deployment choice.

ASSESSMENT

# Project Rubric

Strong projects do not simply chase the highest score. They show controlled experimentation, correct use of evidence, clear cost awareness, and thoughtful discussion of bias, equity, and safety.

SUBMISSION STANDARD

## Evidence over intuition

Support every recommendation with benchmark results, notebook observations, and chart evidence.

| CATEGORY | WEIGHT | WHAT EARNS FULL CREDIT |
|---|---|---|
| **Experimental design** | 25% | Compares settings systematically, changes one variable at a time, and explains why each comparison matters. |
| **Benchmark evidence** | 20% | Uses practice, checkpoint, and final benchmark results accurately; shows what improved and what remained weak. |
| **Cost and observability** | 15% | Correctly interprets token use, latency, retrieved context, and cost implications for real deployment. |
| **Safety and clinical reasoning** | 15% | Identifies unsafe outputs, overconfidence, hallucinations, or chart-grounding failures and proposes guardrails. |
| **Bias and equity analysis** | 15% | Examines language, insurance, SDOH, race/ethnicity, and access-sensitive cases and explains equity risks clearly. |
| **Final recommendation and presentation** | 10% | Presents one defensible configuration for a realistic clinical use case and communicates tradeoffs clearly. |
| **Total** | **100%** | **Recommendation is evidence-based, realistic, and safe enough to discuss as a prototype deployment choice.** |

### Presentation Checklist

- Clinical use case and target user.
- Best configuration and why it won.
- One example where the model performed well.
- One example where the model failed or created risk.
- Token, cost, or latency tradeoff worth noting.
- One bias/equity concern and one mitigation step.

### Avoid These Mistakes

- Changing several settings at once and claiming one caused the improvement.
- Using benchmark scores without checking the actual chart evidence.
- Ignoring cost because the classroom models are free tier.
- Describing equity only in general terms without case-specific analysis.
- Recommending deployment without naming guardrails, monitoring, or human review.

Tool use is allowed, including AI assistants, but your team is responsible for verifying clinical accuracy and citing where AI assisted your process. Use the in-app Team Notebook and reflection as the official record of your work.